

Poznan University of Technology
Institute of Mechanical Technology
Poznan, Poland
dawid.kucharski@put.poznan.pl

June 14, 2026

The Editor
Physica A: Statistical Mechanics and its Applications

Dear Editor,

I am pleased to submit our manuscript entitled “Calibrating the Brody exponent as a quantitative measure of short-range exclusion in 2D spatial point processes” for consideration for publication in Physica A.

Why Physica A? This paper develops a calibrated measurement framework for short-range exclusion in 2D spatial point processes—a statistical-physics methodology demonstrated on manufactured surfaces, prime number embeddings, and interferometric profilometry. The work bridges random matrix theory (Brody distribution) with established spatial point-process statistics, providing the correct 2D reference frame and an empirical calibration curve (Spearman $\rho = 0.988$) that translates the dimensionless Brody parameter into a physically interpretable exclusion radius. The interdisciplinary scope—spanning surface engineering, number theory, and optical metrology—is well-suited to Physica A’s readership.

Key contributions:

1. The 2D complete-spatial-randomness baseline is recalibrated to $\beta = 0.96 \pm 0.15$ (30 realisations, density-stable), correcting the inappropriate 1D Poisson reference $\beta = 0$. A distinct binary-field CSR baseline ($\beta \approx 1.46$ at low fill fraction) is identified, with a decision table provided for practitioners.
2. An empirical β - $r_{\text{excl}}/\langle \text{NN} \rangle$ calibration curve (Spearman $\rho = 0.988$ with effective hard-core radius) translates the dimensionless Brody parameter into a physically interpretable exclusion measure spanning CSR to near-crystalline regimes.
3. A sparse-integer control proves that the elevated $\beta = 2.15$ for 2D prime embeddings is a genuine arithmetic signal ($\Delta\beta = +0.68$ over random-integer control), not a sparsity artefact.
4. The Cantor embedding null result ($\beta = 1.40 < \beta_{\text{Bern}} = 1.47$, TOST $p < 0.01$) demonstrates that 2D exclusion in arithmetic sequences is embedding-created rather than intrinsic—the framework detects both the presence and the absence of exclusion.

5. Density-thinning, embedding-robustness, spatial-permutation, and peak-detection sensitivity controls isolate genuine exclusion from confounding effects across 58 manufactured surfaces (10 materials, 10 processes).

Rigour and reproducibility: All analyses are accompanied by bootstrap confidence intervals, AIC-based model selection, Benjamini–Hochberg multiple-comparison correction, power analysis, and formal equivalence testing (TOST). The complete analysis code, derived results, and generated figures are archived in a public GitHub repository (github.com/dawidkucharski/brody-2d-exclusion). Raw surface and interferometric data are available upon request. Every numerical claim in the manuscript is verifiable against the deposited data and scripts.

Declarations: This manuscript is original, has not been published elsewhere, and is not under consideration by another journal. The author declares no competing interests. This is a single-author submission. The manuscript complies with Physica A’s formatting and ethical guidelines.

Respectfully,

Dawid Kucharski